

# Alphelix GPCR Asset Generation Capability Report

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## Executive Summary

Alphelix Biosciences presents itself as a membrane-protein drug discovery service and platform company, with GPCRs as its flagship technical focus. Based on its public website, the company's core differentiation is upstream enablement: producing stable, purified, ligand-free, native-like GPCR proteins that can be used for antibody discovery, small molecule and peptide screening, structural biology, and functional validation.

The key platform is MegaR, a proprietary GPCR engineering approach that inserts an "M-Protein" into intracellular loops to improve GPCR stability while preserving the transmembrane and extracellular regions. Alphelix claims this enables ligand-free GPCR preparation, purified-protein screening, and cryo-EM structure determination for otherwise difficult GPCR targets.

Alphelix's public materials support a target-to-hit service stack across antibody and small-molecule/peptide asset generation. However, the public evidence is still mainly company-claimed. For diligence, the major open questions are success rates, target-by-target validation, pharmacology preservation versus wild-type GPCRs, hit-to-lead conversion, patent coverage, and customer references.

## Overall Capability Positioning

Alphelix is best viewed as a GPCR and membrane-protein target-to-hit generation platform/CRO rather than a fully integrated therapeutic developer. Its service chain covers:

- GPCR and membrane protein preparation
- Purified ligand-free GPCR product/reagent generation
- Antibody discovery against GPCRs and other membrane proteins
- Small molecule and peptide screening
- Structure determination and AIDD/SBDD support
- Functional validation using protein-based and cell-based assays

The distinctive element across these services is access to stable, bioactive, ligand-free GPCR proteins. This is a meaningful bottleneck in GPCR discovery because many GPCRs are unstable, hard to purify, conformationally heterogeneous, and difficult to use in purified-protein screening or structural biology workflows.

## Core Platform: MegaR

### *Claimed Mechanism*

Alphelix states that MegaR stabilizes GPCRs by integrating a specific protein complex, called M-Protein, into the first, second, or third intracellular loops of GPCRs. The company claims this improves stability while maintaining native conformation in the transmembrane and extracellular regions.

### *Claimed Technical Advantages*

- Stabilization of ligand-free GPCR proteins
- Preservation of native conformation and ligand-binding capability
- Suitability for DEL screening, AS-MS screening, antibody discovery/screening, and ligand affinity measurement
- Increased engineered protein size to over 100 kDa, with greater rigidity for cryo-EM

- Short trial and production timelines

### ***Claimed Scale***

Alphelix states that it has produced more than 110 GPCRs using MegaR. Its product page also claims a purified ligand-free GPCR protein library with 168 completed targets and 165 in progress as of June 2025. The company describes two versions of GPCR proteins: basal-state and agonist-preferring-state proteins.

## **Antibody Asset Generation**

### ***Platform***

Alphelix's antibody discovery platform is MegAb, built on MegaR-enabled membrane protein antigen generation. The platform is designed to address three common pain points in membrane protein and GPCR antibody discovery:

- Lack of highly immunogenic antigens
- Lack of antigens that enhance antibody specificity
- Lack of efficient membrane protein antibody screening methods

### ***Antigen Formats***

Alphelix describes two VLP-based antigen technologies:

- UltraVLP: virus-like particles incorporating immune-enhancing components to boost immune responses to membrane proteins.
- AquaVLP: virus-like particles reconstructed with purified membrane proteins, claimed to achieve more than 90% target-protein representation versus traditional VLPs.

### ***Screening Approach***

The company emphasizes that it can use purified membrane proteins, especially purified GPCRs, for antibody screening. This is differentiated from approaches that rely only on cells or membrane fragments. Alphelix states that it supports traditional hybridoma technology and microdroplet/microfluidic high-throughput sorting.

### ***Service Timeline***

Alphelix lists the following antibody discovery timelines:

- Immunogen preparation: 2-3 weeks
- Animal immunization: 5-8 weeks
- Monoclonal antibody screening and production: 10-12 weeks

### ***Public Case Signals***

Alphelix provides public case references for early discovery of CB1 monoclonal antibodies based on MegAb, and for UltraVLP/AquaVLP improving immunogenicity. The website shows figures, but does not provide full downloadable datasets or detailed antibody profiles in text.

## **Small Molecule and Peptide Asset Generation**

### ***Discovery Model***

For small molecules and peptides, Alphelix positions its advantage around affinity-based screening against stable, ligand-free, bioactive membrane proteins. The company argues that conventional cell-based one-well-one-compound functional assays are labor-intensive, artifact-prone, and limited in throughput. In contrast, purified-protein methods such as

DEL screening and peptide library screening can access larger library diversity if suitable target protein is available.

## ***Partnered Capabilities***

Alphelix states that it has partnered with HitGen for DNA-encoded library screening and with other partners for peptide library access. This suggests Alphelix's internal differentiation is target preparation and validation, while library scale and some screening infrastructure may be partner-enabled.

## ***Service Workflow***

The disclosed small molecule and peptide discovery workflow includes:

1. Protein preparation and activity validation
2. DEL or peptide library screening
3. Hit validation

## ***Service Timeline***

Alphelix lists the following timelines:

- Protein preparation and activity validation: 3-4 weeks
- DEL / peptide library screening: 4-8 weeks
- Hit validation: 4-8 weeks

## ***Public Case Signal***

Alphelix provides a public case reference for early discovery of small molecule agonists for MC4R using MegaR plus DEL, in partnership with HitGen. The public page includes a figure but does not provide full hit structures, affinities, functional potency, selectivity, or medicinal chemistry progression.

# **Structural Biology and AIDD/SBDD**

## ***Structural Biology Positioning***

Alphelix states that MegaR helps systematize ligand-free GPCR and GPCR-antagonist structure elucidation. By increasing molecular weight above 100 kDa and improving rigidity, MegaR-engineered GPCRs are claimed to become more suitable for cryo-EM and structure-based drug design.

## ***Modalities***

Alphelix lists capabilities in:

- Cryo-EM structure determination
- X-ray crystallography, including lipidic cubic phase methods
- Structure-based drug design
- Artificial intelligence-driven drug design

The AIDD/SBDD platform is described as jointly established with DP Technology.

## ***Public Case Signal***

The strongest named case is CCR8. Alphelix states that wild-type CCR8 or CCR8 fused with other partners was difficult to prepare as conformationally homogeneous protein, while MegaR enabled ligand-free CCR8 production. The company also states that:

- A therapeutic antibody interacted with MegaR-CCR8 extracellular regions in ELISA

- Cryo-EM structures were resolved in both ligand-free and antagonist-bound states

Other preparation/validation cases include GPR75, VIP2R, and APJR, assessed with methods such as SPR, MST-TRIC, and fluorescence polarization.

## Functional Validation

Alphelix states that it has developed cell-based functional validation methodologies based on membrane protein expertise. The company lists both protein-based and cell-based validation services.

### Service Timeline

- Protein-based assay: 2-3 weeks
- Cell-based assay: 1-2 weeks

This capability is important because purified-protein binders from antibody, DEL, or peptide campaigns still need to be assessed for biological modulation, pathway effects, and relevance to native receptor function.

## Capability Map

| Stage                    | Alphelix Claimed Capability                                   | Asset Relevance                      |
|--------------------------|---------------------------------------------------------------|--------------------------------------|
| Target preparation       | MegaR-stabilized GPCR and membrane protein production         | Enables difficult target access      |
| Antigen generation       | Purified GPCRs, UltraVLP, AquaVLP                             | Enables antibody discovery           |
| Antibody discovery       | Hybridoma and microdroplet/microfluidic screening             | Generates mAb hits                   |
| Small molecule screening | DEL screening with HitGen and other partners                  | Generates small molecule hits        |
| Peptide screening        | Peptide library screening with partners                       | Generates peptide ligands/modulators |
| Structural biology       | Cryo-EM, X-ray crystallography, MegaR-enabled GPCR structures | Supports SBDD and mechanism work     |
| AIDD/SBDD                | Partnered with DP Technology                                  | Supports design and optimization     |
| Functional validation    | Protein-based and cell-based assays                           | Confirms binding and activity        |

## Assessment of Differentiation

Alphelix's strongest differentiation is likely the upstream target-enablement layer: the ability to make stable, ligand-free, purified GPCR proteins at useful scale. If validated, this capability can unlock multiple downstream asset generation paths:

- Antibodies against difficult extracellular GPCR epitopes
- DEL-derived GPCR binders
- Peptide ligands or modulators
- Cryo-EM structures of ligand-free or antagonist-bound GPCRs
- Affinity and functional assay workflows using purified receptor proteins

The company appears more differentiated in target preparation and screening enablement than in fully integrated drug development. Its small molecule and AIDD/SBDD capabilities appear partner-supported, while its proprietary platform claims sit primarily around MegaR and MegAb.

## Key Diligence Questions

### *Platform Validation*

- For the claimed 110+ GPCRs or 168 completed proteins, what percentage met predefined yield, purity, stability, and activity criteria?
- How does ligand binding compare with wild-type receptor in native membrane or cell systems?
- Does M-Protein insertion alter receptor conformational equilibria, ligand rank order, or signaling bias?
- How many GPCR classes and families have been successfully demonstrated beyond Class A?

### *Antibody Discovery*

- What is the hit rate per target for MegAb campaigns?
- What percentage of antibodies bind native cell-surface receptor rather than only engineered antigen?
- How many functional antagonist, agonist, internalizing, or depleting antibodies have been generated?
- Are there affinity, specificity, cross-reactivity, and epitope-binning datasets?
- Has any antibody advanced into in vivo efficacy or developability testing?

### *Small Molecule and Peptide Discovery*

- For DEL and peptide campaigns, what are the confirmed hit rates after resynthesis?
- What affinities and functional potencies have been achieved?
- Are hits orthosteric, allosteric, agonist, antagonist, inverse agonist, or silent binders?
- What selectivity data exist versus related GPCR family members?
- Does Alphelix conduct medicinal chemistry internally, or only through partners?

### *Structural Biology*

- How many MegaR-GPCR structures have been solved, at what resolutions, and for which targets?
- Are structures deposited, patent-protected, or customer-confidential?
- How well do MegaR-enabled structures match native receptor pharmacology?
- Can MegaR support active-state structures, not just ligand-free or antagonist-bound states?

### *Business and IP*

- What patents cover MegaR, MegAb, UltraVLP, and AquaVLP?
- Are there freedom-to-operate issues around GPCR fusion stabilization methods, VLP antigen display, or DEL screening partnerships?
- What customer references, repeat contracts, or pharma collaborations exist?
- Are assets generated for clients fully client-owned, co-owned, or subject to platform reach-through?

## Preliminary Diligence Conclusion

Alphelix appears to offer a credible and strategically relevant end-to-end discovery service stack for GPCR and membrane protein asset generation, particularly through antibody and small molecule/peptide hit discovery. The most important differentiator is not the downstream screening modality itself, but the ability to generate purified, stable, ligand-free, bioactive GPCR proteins that can feed multiple discovery workflows.

The opportunity is strongest if Alphelix can substantiate its claims with target-by-target data showing preserved pharmacology, reproducible protein production, antibody and DEL hit generation, functional activity, and structure-enabled optimization. Without those data, the company should be viewed as a promising platform with an attractive technical thesis but still requiring deeper validation.

## Sources Reviewed

- Alphelix homepage: <http://alphelix.com>
- MegaR Technology Platform: <http://alphelix.com/platform.html>
- MegAb Membrane Protein Antibody Discovery Platform: <http://alphelix.com/platform1.html>
- Structural Biology Platform: <http://alphelix.com/platform2.html>
- Products: <http://alphelix.com/products.html>
- Service Overview: <http://alphelix.com/cro.html>
- GPCR Preparation: <http://alphelix.com/cro1.html>
- GPCR Preparation Case: <http://alphelix.com/cro1-1.html>
- Antibody Discovery: <http://alphelix.com/cro3.html>
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- Small Molecule and Peptide Discovery Case: <http://alphelix.com/cro4-1.html>
- Structure Determination and AIDD/SBDD: <http://alphelix.com/cro5.html>
- MegaR Cryo-EM Structure Determination: <http://alphelix.com/cro5-3.html>
- Functional Validation: <http://alphelix.com/cro6.html>